

Shopsmart: Your Digital Grocery– Final Project Report

1. INTRODUCTION

1.1 Project Overview

ShopSmart is a MERN-stack based digital grocery shopping platform that connects customers, grocery sellers, and administrators. Users can browse categories, add products to their cart, and place orders. Sellers can manage inventory and orders through a seller dashboard, while admins manage product promotions and platform governance.

1.2 Purpose

To build a real-time, responsive online grocery store with role-based access for customers, sellers, and admins, delivering seamless UI/UX, secure transactions, and efficient backend operations.

2. IDEATION PHASE

2.1 Problem Statement

Local grocery stores and customers often face challenges in digitizing their shopping experience. How might we streamline the grocery shopping process through a simple, accessible, and real-time web platform?

2.2 Empathy Map Canvas

Who are we empathizing with? Customers and local grocery shop owners
What do they see/hear/do/feel?

- See: Disorganized item availability
- Hear: Stock issues, late deliveries
- Do: Visit stores physically or make calls
- Feel: Anxious about freshness, unplanned shopping

2.3 Brainstorming

Grouped Ideas:

- **Customer Experience:** Intuitive UI, categorized items, real-time cart
 - **Seller Experience:** Inventory updates, order management
 - **Admin Tools collaborative brainstorming.:** Manage sellers, promote products
Used sticky notes and Miro for
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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Visit Homepage →

2. Browse Grocery Categories →
3. Add Products to Cart →
4. Login/Signup →
5. Checkout and Place Order →
6. Track Order →
7. Provide Feedback

3.2 Solution Requirement

Functional:

- Role-based login
- Product search, cart management, order placement
- Seller dashboard for stock & orders
- Admin dashboard for promotions

Non-functional:

- Responsive design
- Token-based authentication
- Scalable backend with RESTful APIs

3.3 Data Flow Diagram

Level-0 DFD includes:

- **Actors:** Users, Sellers, Admins
- **Processes:** Auth, Cart, Orders, Product Management, Promotions
- **Stores:** MongoDB collections for users, sellers, products, orders, and carts

3.4 Technology Stack

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Security:** JWT authentication
- **Architecture:** 3-tier MERN-based system

4. PROJECT DESIGN

4.1 Problem Solution Fit

Bridging local grocery stores and customers via a digital, responsive shopping experience with role-specific tools.

4.2 Proposed Solution

Key Features:

- Role-based login (Customer / Seller / Admin)
- Categorized grocery browsing
- Cart and order system
- Seller panel for inventory management
- Admin dashboard for promotions

4.3 Solution Architecture

User Interface (React.js)

↓ API Calls

Application Logic (Node.js + Express)

↓ MongoDB Database

Supports real-time operations, modular structure, and scalability.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Project divided into 4 Sprints
- Each sprint delivers core functionality
- Used story points, burndown charts, and velocity tracking to manage progress

6. FUNCTIONAL AND PERFORMANCE TESTING

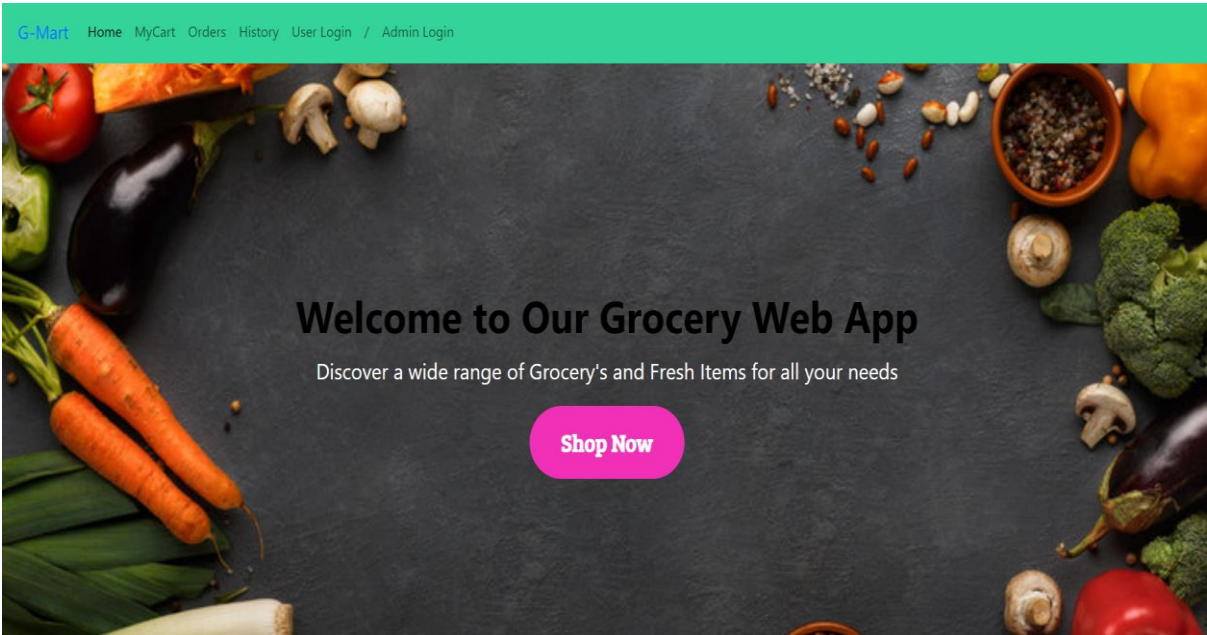
6.1 Performance Testing (UAT)

- Secure registration/login for customers, sellers, admins
- Product cart and checkout tested
- Inventory updates from seller side
- Role access verification
 - ✓ All core features tested and validated

7. RESULTS

7.1 Output Screenshots

Home Page with flight search



Login form



Login

Email

Enter email

Password

Enter password

Login

Don't have an account? [Sign Up](#)

Items Page:

[G-Mart](#) [Home](#) [MyCart](#) [Orders](#) [History](#) [Logout](#)

Search By Product Name

Search by product name

Filter By Category

all

Products



Apple
\$200

Buy NowAdd to Cart



orange
\$120

Buy NowAdd to Cart



Milk
\$80

Buy NowAdd to Cart



Cashew
\$800

Buy NowAdd to Cart



Chicken
\$250

Buy NowAdd to Cart

My Cart:-

[G-Mart](#) [Home](#) [MyCart](#) [Orders](#) [History](#) [Logout](#)

My Cart



Apple
\$200

Remove from CartBuy this



Milk
\$80

Remove from CartBuy this



Chicken
\$250

Remove from CartBuy this

My Orders Page:-

[G-Mart](#) [Home](#) [MyCart](#) [Orders](#) [History](#) [Logout](#)

My Orders

Order ID: 6614b085c30b51d3c700f20a
Name: syed arshad
Phone: 9505221870
Date: 2024-04-09T03:05:41.563Z
Price: 400
Status: Pending
Payment Method: credit

My History Page:-

[G-Mart](#) [Home](#) [MyCart](#) [Orders](#) [History](#) [Logout](#)

My History

Order ID: 6614a8ddc30b51d3c700f1b4

Name: syed arshad

Phone: 9505221870

Date: 2024-04-09T02:36:47.498Z

Price: 400

Status: Delivered

Payment Method: debit

Place Order Page:-

[G-Mart](#) [Home](#) [MyCart](#) [Orders](#) [History](#) [Logout](#)

Order Details

First Name:

Last Name:

Phone:

Quantity:

Address:

Payment Method:

Cash on Delivery (COD)

Submit

localhost:3000/login

Users Page:-

Grocery Web App

DashboardUsersProductsAdd productOrdersLogout

Users

sl/no	UserId	User name	Email	Operation
1	6614a859c30b51d3c700f1a0	syed	syed@gmail.com	view

Add Product page:-

Grocery Web App

DashboardUsersProductsAdd productOrdersLogout

Add Product

Product Name

Enter product name

Rating

Enter product rating

Price

Enter product price

Image URL

Enter image URL

Category

Select Category

Count in Stock

Enter count in stock

Description

Enter product description

Add Product

Admin Orders Page:-

Grocery Web App

DashboardUsersProductsAdd productOrdersLogout

Orders

Order ID: 6614a8ddc30b51d3c700f1b4

Fullname: syed arshad

Phone: 9505221870

Product ID: 660e84641c659d0b2b10e135

Quantity: 2

Total price: 400

Payment Method: debit

Address: hyderabad

Created At: 2024-04-09T02:36:47.498Z

Status: Delivered

Delivered

Order ID: 6614b085c30b51d3c700f20a

Fullname: syed arshad

Phone: 9505221870

Product ID: 660e84641c659d0b2b10e135

Quantity: 2

Total price: 400

8. ADVANTAGES & DISADVANTAGES

✓ Advantages

- Real-time grocery ordering
- Role-specific dashboards
- Responsive UI on all devices
- Secure authentication using JWT
- Admin control over featured products

✗ Disadvantages

- No native mobile app yet
- Payment gateway integration pending (demo phase)

9. CONCLUSION

ShopSmart successfully simulates a full-featured online grocery platform using MERN technologies. It incorporates secure login, live cart management, seller dashboards, and an admin panel for governance.

10. FUTURE SCOPE

- Integrate real payment gateways (e.g., Razorpay, Stripe)
 - Develop a delivery tracking system
 - Create a mobile version using React Native
 - Enable customer reviews and ratings
 - Add sales analytics for admin/seller dashboards
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11. APPENDIX

- **Source Code:** <https://github.com/simhadri-nandini17/Shopsmart-your-digital-grocery-store-experience>
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